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GGIS 403

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May 11, 2026

Spatial Inequality in Access to Special Education Services in Illinois

Introduction and Research Question

Special education services lie at the heart of public education policy. They determine whether students with disabilities access the supports, accommodations, and inclusive learning environments they need. Another way to think about access to special education services is geographically. Schools reside in counties, and counties vary in their population size, school capacity, rurality, transportation distance, staffing levels, and administrative capacity at the local level. For that reason, two students with similar disabilities and special education needs might find themselves in vastly different school service environments based on where they live.

This project analyzes spatial variation in select special education indicators across Illinois counties. I am interested in this topic because it allows me to blend my education interests with the GIS skills I learned in this course. The research questions are as follows: How are special education needs distributed across Illinois counties, as measured by the percent of students receiving Individualized Education Program (IEP) at the school level? Are any Illinois counties characterized by both high average IEP enrollment and small school systems? If so, these counties might deserve further study. This project is not designed to prove that any given county is necessarily underserved. Rather, it generates exploratory evidence of county level variation in IEP enrollment.

Review of Existing Work

Previous research shows that special education identification is shaped by more than student need alone. Morgan et al. (2015) found that minority students were disproportionately underrepresented in special education across several disability conditions after controlling for student,

family, and school-level factors. This study is helpful for my project because it shows that IEP enrollment patterns are influenced by institutional and contextual conditions, not only by the actual number of students who need services. Therefore, county-level differences in IEP enrollment should be interpreted.

Other research focuses more directly on rural special education access. McCabe and Ruppar (2023) reviewed rural inclusive education for students with disabilities in the US and found that rural schools may benefit from strong community relationships but may also face challenges related to staffing, service availability, and limited resources. Smith and Krieg (2022) also studied rural Midwestern parents' experiences with the IEP process and found that families faced barriers to involvement in special education planning. Through these studies, I learned that many Illinois counties are rural or have small school systems, which support my decision to examine counties that combine higher IEP enrollment with smaller school counts.

Geographic and GIS education research informs this project as well. Drescher et al. (2022) examined the geography of rural educational opportunity and argued that place matters when studying educational access and inequality. While Abulibdeh et al. (2024) used GIS to analyze the spatial distribution and accessibility of public or private schools in Qatar. Although their study is not about special education nor Illinois, it demonstrates how GIS can reveal spatial patterns in educational services. Together, these studies show that education access is not only a school policy issue, but also a geographic issue.

My project builds on this existing work by using public school-level data from Illinois and mapping IEP enrollment patterns at the county level. In specific, my project uses publicly available ISBE data and county boundaries. Indeed, it is more limited than prior research because it does not measure disability category with student records nor service quality by parent or staff interviews. However, it contributes an exploratory GIS view of where average IEP enrollment is higher or lower across Illinois counties.

Dataset and Variables

The datasets used in this project come from public sources. Dataset 1 is the 2025 Illinois Report Card Public Data Set from the Illinois State Board of Education (ISBE). The sheet *General* was selected, because it contained school name, county location, and variable *% Student Enrollment - IEP*. Dataset 2 is the 2023 United States Census Bureau TIGER/Line county shapefile. The counties shapefile was filtered down to counties within Illinois by STATEFP = 17.

The main variable of interest is *iep_percent*, which represents the percentage of student enrollment with IEP at each school. Other variables were created during aggregation to describe county-level patterns. The variables used in the analysis are listed in Table 1. There were 99 missing values for IEP percentage, making up 2.59% of the cleaned school-level data. I did not replace missing values with zero, because missing data does not necessarily mean that a school had no students with IEP, maybe due to not reporting the information. Moreover, administrative records such as *State of Illinois* and *Dept of Corrections* were excluded from interpretation.

Variable	**Description**
iep_percent	Percent of student enrollment with IEP at each school
school_count	Number of school records in each county
valid_iep_count	Number of schools with non-missing IEP percentage values
schools_with_iep	Number of schools reporting positive IEP enrollment percentage
mean_iep_percent	Average school IEP percentage in each county
median_iep_percent	Median school IEP percentage in each county

pct_valid_iep_data	Percent of school records in each county with valid IEP data
possible_high_need_small_system	Exploratory flag for counties with above median mean IEP percentage and below median school count

Table 1. Variables Used in the Analysis

Methods

First, I used pandas for data cleaning. I selected the General sheet from the ISBE workbook. Then, I renamed columns, cleaned county names, converted the IEP percentage variable to numeric format, and checked missing values. Missing values were excluded from mean and median calculations. Second, I used attribute aggregation with groupby to summarize school-level data at the county level, because the final spatial unit of analysis was county, not individual schools. The aggregation calculated the interested variables listed above. Third, I used GeoPandas to load the US Census TIGER/Line county shapefile, then filter it to Illinois counties, and project the county boundaries to EPSG:26916. Fourth, I performed an attribute join between the county level ISBE summary table and Illinois county polygons using a cleaned county key. Finally, I used descriptive charts and choropleth maps to interpret the results. These visualizations included a histogram, bar charts, scatter plot, county maps, and an exploratory high IEP need and small school system flag.

Results and Interpretation

The cleaned school-level data was largely complete. The average school level percentage was approximately 17%, while the median percentage was about 15.8 %. Values ranged from 2.9 % to 95.8 %. Because of the relatively low missing rate, aggregating to the county level was a valid method of exploration. The spatial join was successful: all 102 Illinois county polygons received ISBE summary values, and there were zero missing joins. See from the histogram below that county average IEP percentages cluster towards the center of the range. Most counties have between roughly 15 % to

21 % of students enrolled in IEP programs. There are some counties that are significantly higher or lower. Although most counties are not extreme outliers, Illinois counties do not appear to report identical enrollment patterns in special education programs.

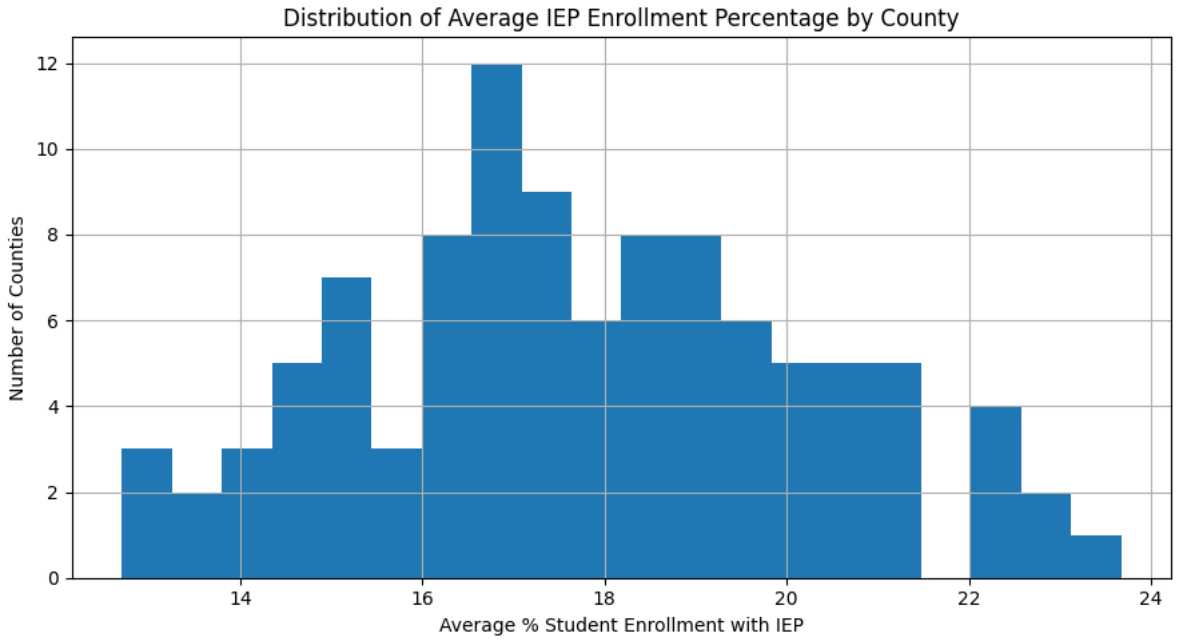


Figure 1. Distribution of average IEP enrollment percentage by county.

Specifically, Bond, Pope, Hamilton, Fulton, Gallatin, Livingston, Marion, Clark, Bureau and Mason Counties were in the top 10 counties for highest average IEP enrollment percentages. Bond had the highest county average percentage of interpreted counties at approximately 23.67 %. Both Pope and Hamilton Counties were over 22 %. These counties do not necessarily imply services are worse than in others, but that is where schools reported a higher average percentage of students with IEP.

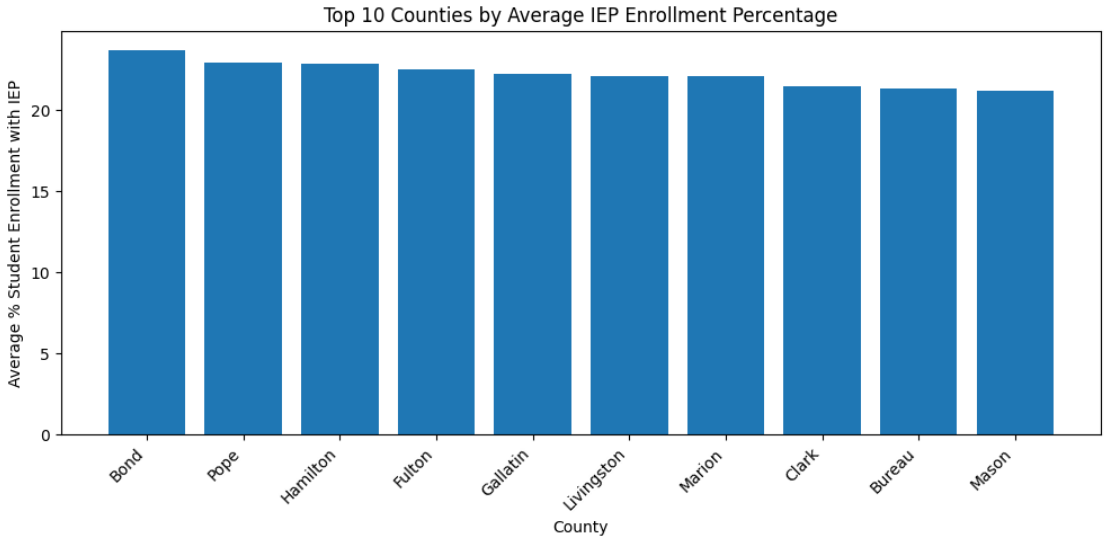


Figure 2. Top 10 counties by average IEP enrollment percentage.

In contrast, Monroe, Lawrence, Woodford, Edwards, Hancock, Jersey, Cumberland, Richland, Calhoun, and Lee were among the lower average counties. Monroe had the lowest county average at just over 12.7%. The difference between Monroe and Bond exceeds ten percentage points, enough to indicate that useful county-level variation exists, but county averages may conceal variation between schools within counties.

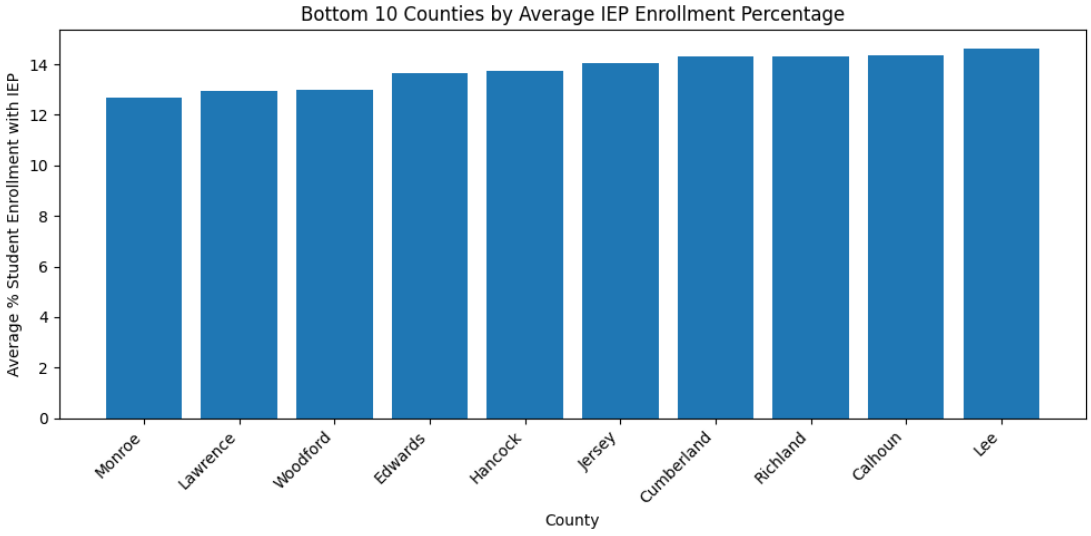


Figure 3. Bottom 10 Counties by Average IEP Enrollment Percentage

Based on the following scatter plot, there was a really weak correlation between school count and average IEP %. The correlation coefficient was around -0.071, which is near zero. Counties that

had higher numbers of schools were not clearly distinguished by higher or lower average IEP enrollment. So that, school count and average IEP % should be viewed as separate attributes of county school districts.

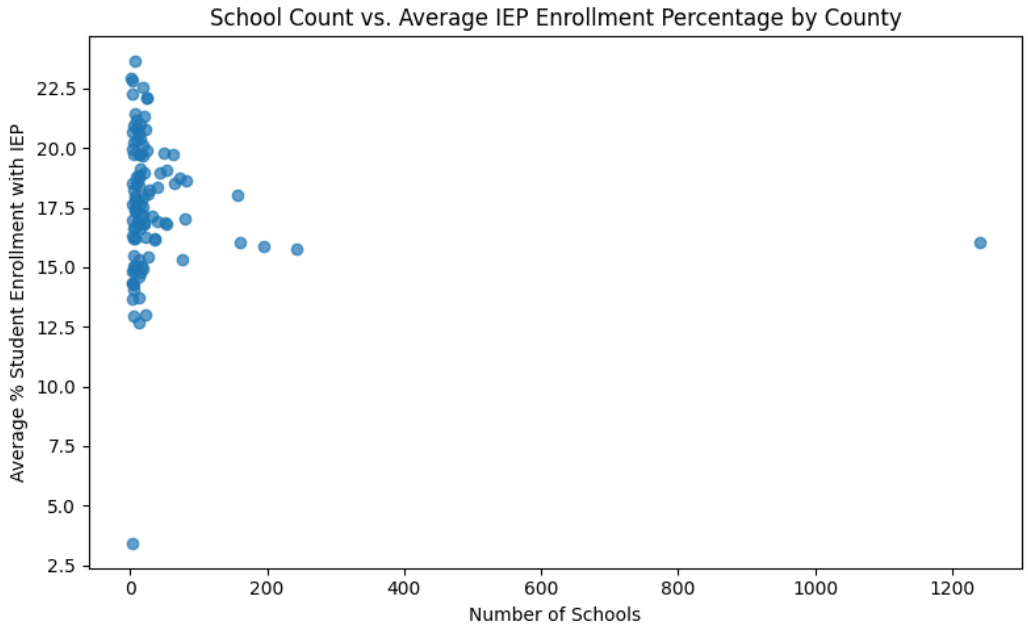


Figure 4. School count versus average IEP enrollment percentage by county.

The choropleth map below visually displays the clearest spatial evidence supporting the claim that mean IEP enrollment varies by county in Illinois. The darkest grouping on the map ranges from counties with mean IEP percentages of about 19.95% - 23.67%. The lightest grouping contains counties ranging from about 12.70% - 15.57%. There is visible distance between counties on the low end of IEP enrollment and those on the high end. Counties with higher percentages of students with IEPs spans various areas of western, central, and southern Illinois while counties with lower averages also dot northern and eastern Illinois.

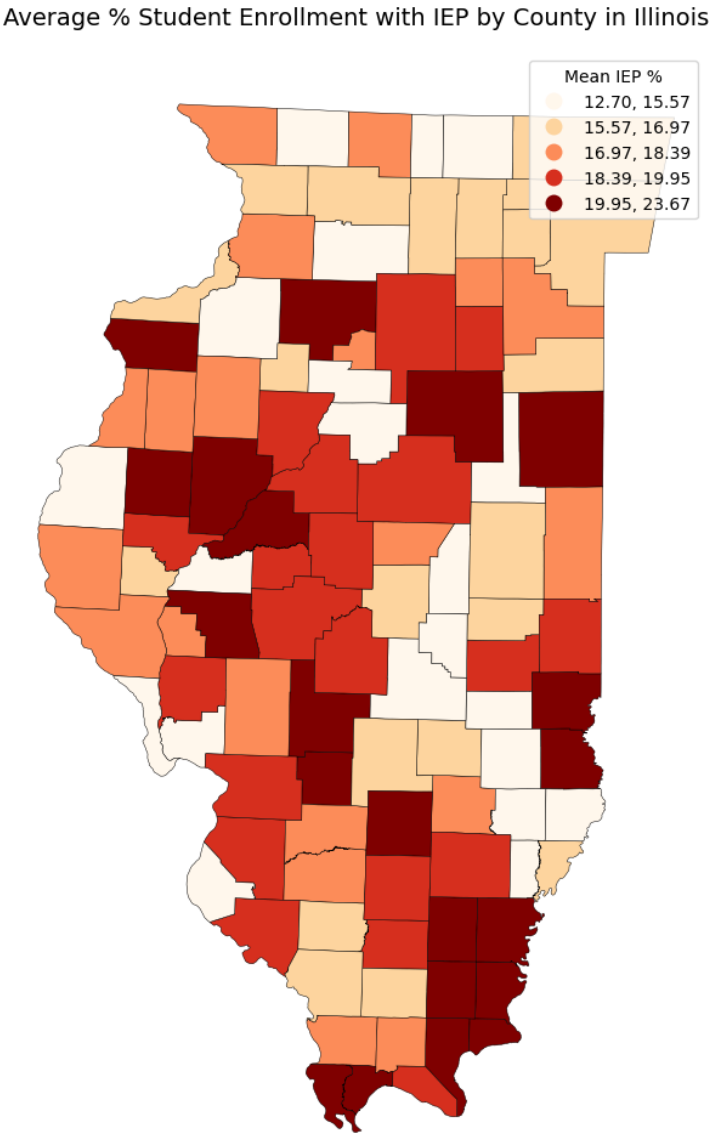


Figure 5. Average percent student enrollment with IEP by county in Illinois.

The final exploratory analysis identified counties that combined high IEP enrollment with small school systems. I used a median average IEP threshold of about 17.63 % and a median school count threshold of 13 schools. There were 26 counties met both conditions, including Bond, Pope, Hamilton, Gallatin, Clark, Mason, Mercer, White, Pulaski, McDonough, Crawford, Alexander, Hardin, Greene, and Coles. These counties are not proven underserved areas. They are locations where high IEP enrollment and smaller school systems overlap, so they may warrant further study of the service quality using staffing, funding or service location data.

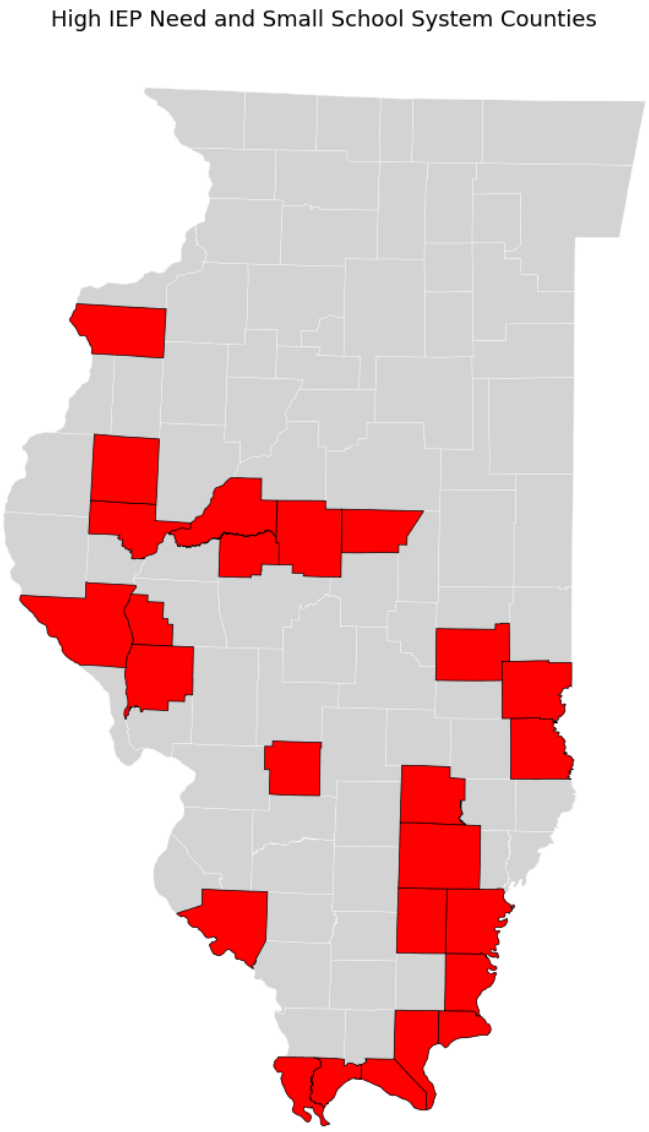


Figure 6. High IEP need and small school system counties.

Conclusion and Key Findings

One finding from the project is that special education-related enrollment does not evenly distributed across the Illinois counties. The strongest evidence for this comes from the clear patterns displayed on Figure 5, depicting average IEP enrollment percentage. There is a noticeable gap between ranges of values from high IEP and low IEP counties, which shows that Illinois counties do not all report the same level of IEP enrollment. Counties with higher average IEP percentages cluster more

regularly in parts of western, central, and southern Illinois. On the flip side, counties with lower average percentages cluster regularly in some northern and eastern areas of the state.

Bar charts (Figure 1, 2, 3) depicting average IEP enrollment leads to the same conclusion. For example, some of the counties with highest average IEP enrollment were Bond, Pope, Hamilton, Fulton and Gallatin counties. The difference between the highest IEP and lowest IEP counties is over ten percentage points, which concludes that there is meaningful variation at the county-level.

Another important finding can be made from Figure 6, the high IEP need and small school system map. If a county had an average IEP percentage at or above the median and a school count at or below the median, then it was flagged by the conditional formatting. With these parameters, 26 counties were flagged: Bond, Pope, Hamilton, Gallatin, Clark, Mason, Mercer, White, Pulaski, McDonough, Crawford, Alexander, Hardin, Greene, and Coles counties. These counties could potentially be under higher pressure than their neighbors, because they exhibit both high reported levels of IEP enrollment and low school counts. Small schools systems may not have enough staff or program options to accommodate students with disabilities. Distance could also be a concern if specialized services are only hosted at a single location within the county. Schools would then be responsible for transporting students to these locations and other students may not have programs hosted at their own school. It could be argued that these highlighted counties are the best candidates for future investigation into whether true special education service lacking exist.

Of course, this conclusion comes with limitations. A county's IEP enrollment percentage does not definitively mean that students have access to services, or the services are better or worse than other areas. A high IEP percentage could be reflective of many different factors, including but not limited to: greater student need, better identification practices, or availability of specialty programs. Another limitation of this study is that data was only analyzed at the county level. This could be problematic, because it conceals variation that occurs between schools or districts within the same county. This ties into the Modifiable Areal Unit Problem, because results of the analysis would

probably look different if school points were used instead of ZIP codes. School coordinates may be utilized in further analysis to discover correlations between special education enrollment and school size of staff, number of programs offered at a school, or distance away from specialty programs. Perhaps school district borders can be included to see if there are differences in enrollment in the same county. For instance, funding, transportation distance to specialty services, main disability categories of student populations, and student/family experiences could also be discussed. In summary, this project finds that there is exploratory GIS evidence to suggest special education enrollment varies from counties across Illinois. Furthermore, counties with high special education need percentages and small school systems seem to be the target for future studies attempting to locate true service inequality.

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